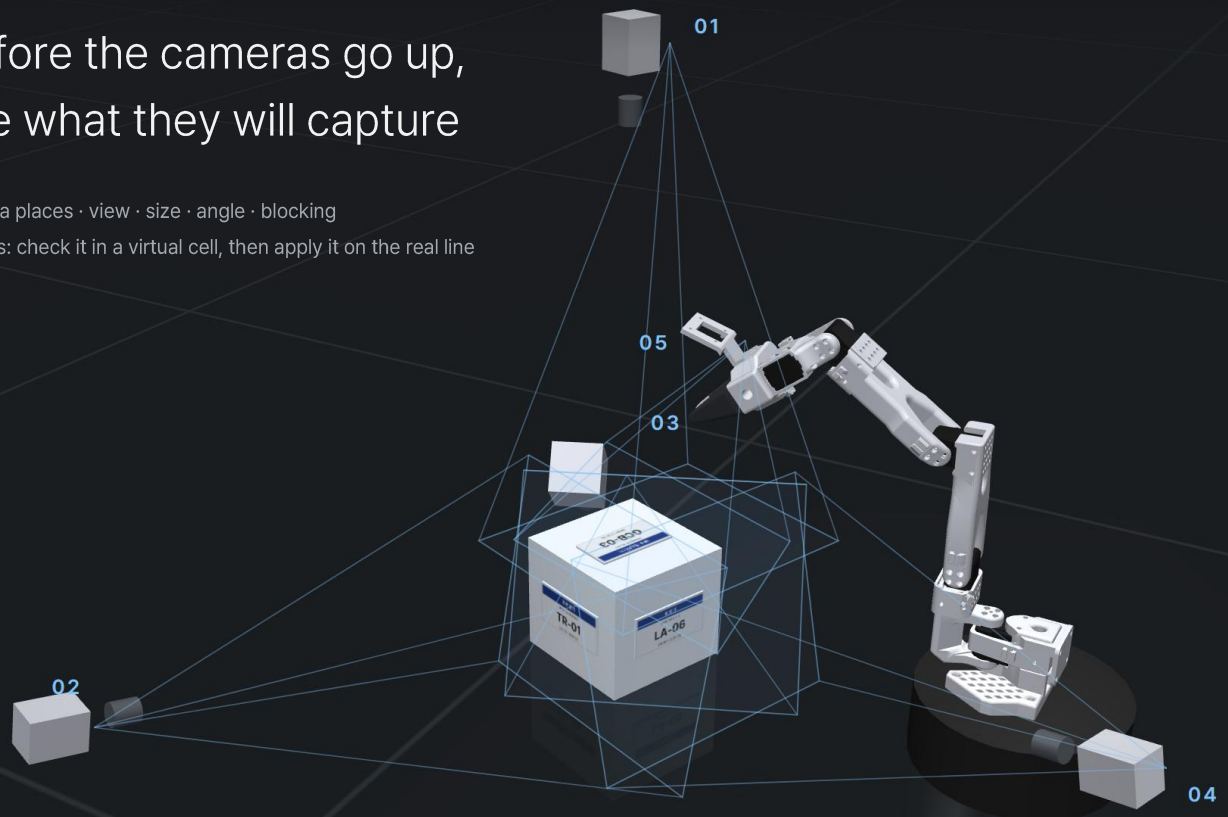
VALIDATE BEFORE INSTALLATION · OPTICAL INSPECTION SIMULATION

IntLearn Kinesis

Before the cameras go up, see what they will capture

Camera places · view · size · angle · blocking
8 steps: check it in a virtual cell, then apply it on the real line





01 TOP | 02 FRONT | 03 CORNER | 04 SIDE | 05 WRIST

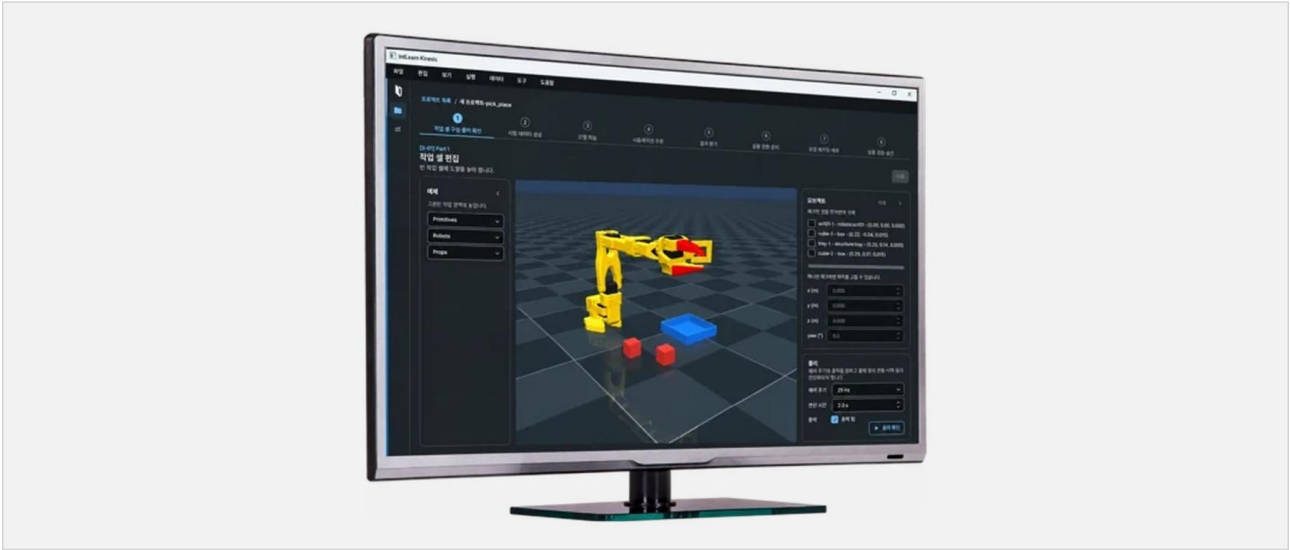
IntLearn Kinesis example 'OCR (xArm7)' · five camera places around the label cube · below, what each one captures

02 — OVERVIEW

Place cameras in a virtual cell, and check each view in advance

Stack robots, conveyors and cameras like blocks into a virtual cell, train on synthetic data instead of waiting for defects, and check it before it goes on the real line. Here, the inspection cameras go in first.

FIG. 01 SOFTWARE



IntLearn Kinesis software screen · example virtual cell (Korean UI)

8 steps
Virtual to real line
From cell to real-world approval

4 checks
Is the label captured?
View · size · angle · blocking

2 ways
To mount a camera
Fixed in place · on a robot

2026
New product
Launched in 2026

Key features From the product page

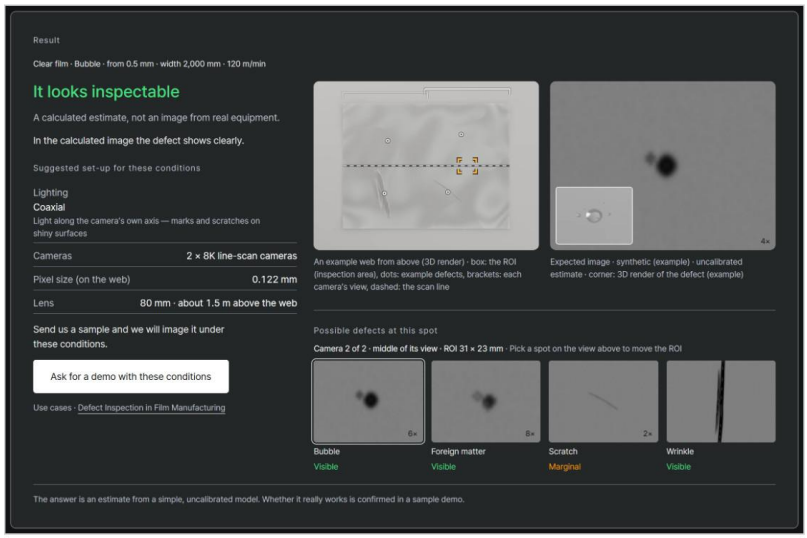
01	Build the virtual cell	Stack robots, conveyors and cameras into a cell · four camera views
02	Synthetic data	Make defect images instead of waiting for them
03	Training · virtual judging	Check performance before going on site
04	Real-world transfer · approval	Take what was validated virtually to the real line
05	Patent filed	Sim-to-Real virtual validation (check virtually, then apply on the real line)

03 — BEFORE INSTALL · OPTICS LAB

Before the cameras go up, check two things first

In the Optics Lab on intelliz.net, pick a material or part and the defect you worry about: it calculates whether line scan (moving material, one line at a time) or area scan (one frame per part) can inspect it, and with which light, camera and lens. IntLearn Kinesis then places that camera in the virtual cell and checks that the target is really captured there.

FIG. 02



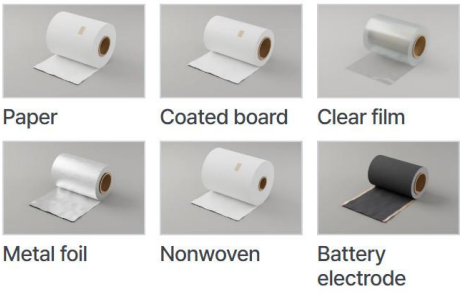
Optics Lab screen · line-scan result example (clear film · bubble · from 0.5 mm · width 2,000 mm · 120 m/min)

A calculated estimate, not an image from real equipment. The answer is an estimate from a simple, uncalibrated model; whether it really works is confirmed in a sample demo.

OPTICS LAB · INTELLIZ.NET

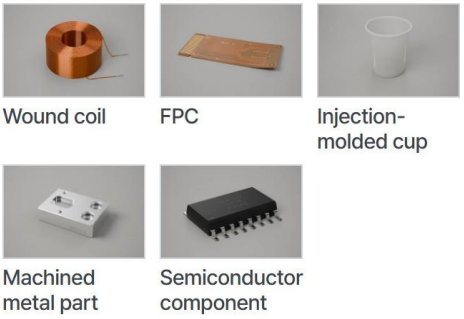
LINE SCAN · IntScan Pro

Continuous materials



AREA SCAN · IntSight

Individual parts



01 OPTICS LAB

Can the defect be seen?

Pick a material or part and a defect: it calculates the set-up and an expected image.

02 KINESIS VIRTUAL CELL

Is it captured from there?

Place the camera in the virtual cell; four checks (view, size, angle, blocking) judge the capture.

03 SAMPLE DEMO

Check it for real

We image your sample on real equipment to confirm it, then move on to the real line.

Optics Lab · intelliz.net

Can Line Scan or Area Scan Inspect It?

intelliz.net/en/solutions/optics-lab

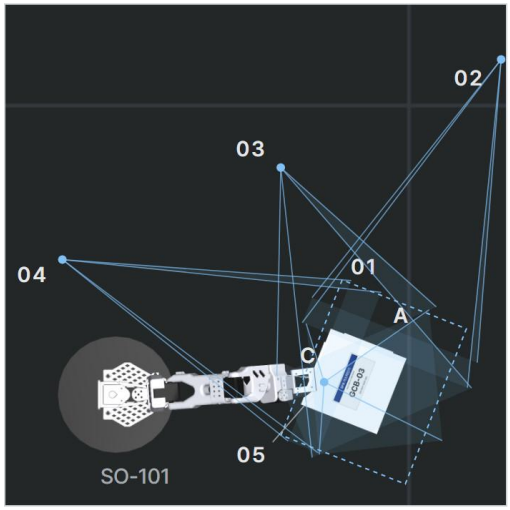
A web tool you use right on the website, separately from IntLearn Kinesis.

04 — CAMERA PLACEMENT

Move the camera around, compare what each one sees

In the work cell screen, fix a camera in place or mount it on a robot, then set its position, height, direction, tilt and view angle. You see each view right away. Below, five cameras surround a three-label cube.

FIG. 03



ONE CELL · FIVE CAMERA PLACES

CAMERA	Label A	Label B	Label C
01 Top Fixed · view angle 25.8°	Not visible	**Pass**	Not visible
02 Front Fixed · view angle 25.8°	**Pass**	Too small	Not visible
03 Corner Fixed · view angle 25.8°	Too oblique	Too oblique	Too oblique
04 Side Fixed · view angle 25.8°	Not visible	Too oblique	**Pass**
05 Wrist SO-101 wrist · 48.5°	Not visible	**Pass**	Too oblique

Not visible = the label faces away from the camera (angle over 90°)



01 TOP 02 FRONT 03 CORNER 04 SIDE 05 WRIST

IntLearn Kinesis example 'OCR (xArm7)' label cube · plan from above (left) and what each camera captures (below) · 01–04 fixed cameras, 05 SO-101 wrist camera

Camera placement Work cell screen

01	Fixed camera	Fix it in one place and set position, direction, tilt and view angle
02	Robot camera	Moves with the robot it is mounted on · built-in wrist cameras are used as is
03	Preview	A small view of what each placed camera will capture, at once
04	Camera list	The placed cameras make the list of views for AI training and for checking

Fixed cameras 01–04 in the figure use the real camera module's 1280:1024 frame and 25.8° vertical view angle. Judgments are the values computed by the product's label-capture check.

05 — CAPTURE CHECK

Will the label be readable? Four checks decide

Four checks decide whether the camera sees the label well enough to read it: is it inside the frame (view), large enough (size), not too oblique (angle), and not blocked by something in front (blocking)? This is not text recognition (OCR). It is calculated only from where the camera and the label are, so it is checked again and again while the robot moves, without drawing the image.

FIG. 04 LABEL A · FIVE PLACES · FOUR CONDITIONS



01 Pass		02 Too far		03 Too oblique		04 At the edge		05 Blocked in front	
In view	pass	In view	pass	In view	pass	**Out of view**	**fail**	In view	pass
Size 30%	pass	**Size 11%**	**fail**	Size 30%	pass	Size 31%	pass	Size 30%	pass
Angle 11°	pass	Angle 11°	pass	**Angle 58°**	**fail**	Angle 11°	pass	Angle 11°	pass
Not blocked	pass	Not blocked	pass	Not blocked	pass	Not blocked	pass	**Blocked**	**fail**

Label A (TR-01) seen by the same camera from five places · outline = the label · bold line = the condition that blocks

Check conditions IntLearn Kinesis label-capture check

01	View	Label corners 5 % inside the frame edge · 1280:1024 frame as on the real camera
02	Size	The label's long side is at least 25 % of the frame width
03	Angle	The label faces the camera within 40° of its line of sight
04	Blocking	Sight lines from the camera to the label's center and four corners hit nothing else first

The check starts with the view and reports the first condition that fails as the reason. For comparison, the figure computes each of the four conditions on its own. In the label-capture task, holding a passing pose for 1 second records the label as captured.

Capture it in the virtual cell before you install.

Move the camera, check again, settle on the right place, then install.

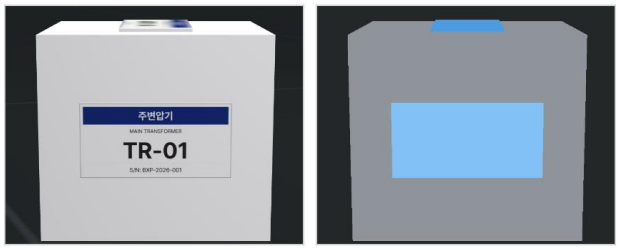
06 — VISIBILITY · FIELD CAMERAS

Was it seen?
Count the pixels

The virtual cell knows which object every point (pixel) of the image belongs to. So you can measure how much of the frame the target fills and how much an object in front hides it. These values are the answer the simulator already knows, not a recognition result from the image, and lighting and material play no part.

FIG. 05 VISIBILITY · RGB / SEGMENTATION

Nothing in front CLEAR



Label A **71,379 px** · 5.45 % of the frame

Capture check **Pass**

Post in front BLOCKED



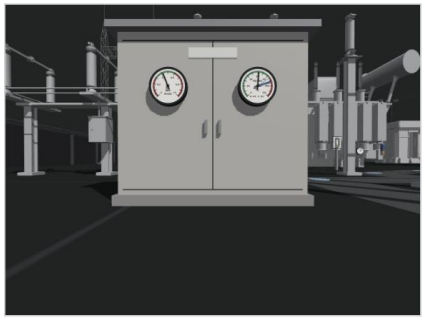
Label A **38,138 px** · 2.91 % of the frame

Capture check **Blocked**

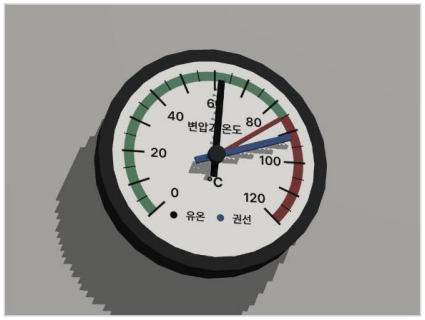
Left: the camera view. Right: the virtual cell colors each pixel by its object · blue = label A

The same goes for a patrol robot. At each patrol point it measures whether the front camera saw the target, zooms in with a PTZ camera (one that pans, tilts and zooms), and makes a virtual thermal image.

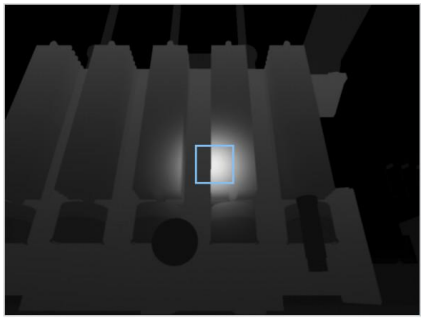
FIG. 06 FIELD CAMERAS · PTZ · THERMAL (SIMULATED)



Front camera at the gauge box



PTZ close-up dual-needle gauge



Virtual thermal whiter = hotter

IntLearn Kinesis example 'Outdoor substation patrol' · the thermal view is simulated; only its peak, 96 °C, is a value set in the example

Light on real samples, placement in the virtual cell.

Check what light changes in the Optics Lab and a sample demo; settle where each camera looks before you install.

07 — EVIDENCE · PLATFORM

Cameras chosen virtually, carried to the real line

Camera places chosen in the cell go through data, training, judging and evaluation to approval on the real line. There, IntCloud Optic, linked through the platform, takes over the optical inspection.

8 steps Virtual to real line

Sim-to-Real · from cell to real-world approval

VIRTUAL virtual cell · 01–05 | **REAL** real line · 06–08

01	02	03	04	05	06	07	08
Cell	**Data**	**Training**	**Inference**	**Evaluation**	**Transfer prep**	**Deployment**	**Real-world approval**
Stack robots and cameras into a cell	Defect images, without waiting	AI trained on virtual images	AI judges in the virtual cell	Checked before going on site	Ready to move to the real line	Deployed to the real line	Validated and approved on the line

IntLearn Kinesis Sim-to-Real in 8 steps · check it in a virtual cell, then apply it on the real line

Products linked on the platform



IntCloud Optic

An AI optical inspection module that captures defects so they can be seen



IntLearn Pro

Deep learning inspection software that finds defects like an expert, with 12 engines and modules, the most in Korea, and LLM integration



IntWise New in 2026

Operate by speaking; the company keeps the know-how

Product videos

Point your phone camera at a QR code

Main · 70 s
What Does a Robot Dog Inspect at a Substation?

Series · 40 s
Build a Virtual Substation for Robot Dog Patrols

Series · 35 s
Three Robot Dog Models, One Virtual Workspace

YouTube @intelliz
More on our channel



IntLearn Kinesis example ‘OCR (xArm7)’ · five camera places and what each one sees

08 — CONTACT · NEXT STEPS

Considering a similar inspection?

Including the following in your inquiry helps us prepare our answer.

01	What you inspect: material, shape and size	02	The defects to find and your pass/fail criteria
03	Line speed and how products are conveyed	04	Space available for cameras and lighting
05	Where results need to go (PLC, reject unit, etc.)		

Next steps

Links on intelliz.net

Optics Lab	Can Line Scan or Area Scan Inspect It?	intelliz.net/en/solutions/optics-lab
Guide	AI Visual Inspection vs Rule-Based Machine Vision	intelliz.net/en/guides/ai-visual-inspection-basics
Getting ready	AI Inspection Adoption Checklist	intelliz.net/en/tools/inspection-checklist
FAQ	Can performance be checked before installation?	intelliz.net/en/company/faq#before-install

CONTACT

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Product page

Try it

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Main Office #305, 63, Buldang 36-gil, Seobuk-gu, Cheonan-si, Chungcheongnam-do, Republic of Korea
Manufacturing Lab B-403, Hyundai Silicon Alley Dongtan, 150, Dongtanyeongcheon-ro, Hwaseong-si, Gyeonggi-do, Republic of Korea
Japan distributor BIT DEVICE Co., Ltd., 6-3-24-L430 Nishinakajima, Yodogawa-ku, Osaka, Japan

INTELLIZ

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Renders: brochure scenes from the IntLearn Kinesis assembly engine (colors; camera markers 60 %).
Judgments and pixel counts come from the product’s capture check and visibility measure, not recognition.
The example’s xArm7 is rendered as SO-101 · SO-101 model: MuJoCo Menagerie (Apache-2.0)
Patrol views: substation structures kevinhalim9 · Gazebo Fuel (CC BY 4.0) · thermal is simulated
Optics Lab screen: intelliz.net/en capture (2026-11-03); results are calculated estimates.
The examples’ labels and gauge faces keep their Korean text, as in the product.